An Additive-Leakage Compiler for Signed Gain and Endpoint Budgets

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Abstract

The preceding four-packet analysis identifies a normalized deficit of the reassembled energy as the exact input needed for a power gain, but it leaves open how a non-multiplicative error should be charged. We prove a conditional two-term compiler. If $D \geq dX^a$ and

$$G \leq BX^{-\gamma}D + \ell X^{a-\delta},$$

then

$$\frac{G}{D} \leq BX^{-\gamma} + \frac{\ell}{d}X^{-\delta} \quad \text{and} \quad \frac{D}{G} \geq \left(BX^{-\gamma} + \frac{\ell}{d}X^{-\delta}\right)^{-1}.$$

The usual one-exponent summary has exponent $\kappa = \min(\gamma, \delta)$ and constant $B + \ell/d$. Combining this with the exact signed-margin identity gives the effective loss $\max(0, \eta_D - \kappa/2)$ and the strict endpoint test $\sigma - \eta_{\text{eff}} > 1/400$. An equality family proves that the two-term denominator is sharp under the stated hypotheses; in particular, a slower additive leakage term cannot be silently assigned the main-term exponent. Six exact rational budget fixtures, four margin fixtures, four endpoint fixtures, and a twelve-row transfer from the TPC-279 parent are independently certified. No arithmetic L^2 estimate or twin-prime conclusion is claimed.

1 The source interface

Let $X \geq 1$ be the scale, let $D > 0$ denote the packet-diagonal source energy, and let $G \geq 0$ denote the energy after signed reassembly. The previous structural criterion uses the ratio $q = G/D$ and gain $r = D/G$ when $G > 0$; see the exact four-packet identities in the TPC-279 release [1]. In an actual analytic estimate it is natural to separate a main term, proportional to D, from a leakage term which is bounded only in absolute scale. We therefore assume

$$D \geq dX^a, \qquad G \leq BX^{-\gamma}D + \ell X^{a-\delta}, \tag{1}$$

where $d > 0$ and $B, \ell, \gamma, \delta \geq 0$. The exponent a is deliberately left symbolic; the intended source interface may have $a = 5/3$.

The distinction between the two terms matters. A multiplicative estimate can be normalized by D directly, whereas an additive estimate must first use the lower bound on D. The next theorem records exactly that operation.

2 The two-term compiler

Theorem 1 (two-term additive-leakage compiler)**.** *Assume* (1) *and set*

$$c_\ell = \frac{\ell}{d}, \qquad \kappa = \min(\gamma, \delta), \qquad C = B + c_\ell.$$

If $C > 0$, then

$$q = \frac{G}{D} \le BX^{-\gamma} + c_\ell X^{-\delta}, \tag{2}$$

$$q \le CX^{-\kappa}. \tag{3}$$

Consequently, if $G > 0$,

$$r = \frac{D}{G} \ge \left(BX^{-\gamma} + c_\ell X^{-\delta}\right)^{-1} \tag{4}$$

$$\ge C^{-1} X^{\kappa}. \tag{5}$$

If $C = 0$, the hypotheses force $G = 0$.

Proof. Divide the second inequality in (1) by D and use $D^{-1} \le d^{-1}X^{-a}$ for the additive term. This gives (2). Since $X \ge 1$ and both exponents are at least κ, $X^{-\gamma}, X^{-\delta} \le X^{-\kappa}$, proving (3). Positive reciprocal reversal gives (4) and (5). If $C = 0$, both terms on the right of (1) vanish. □

The two-term form is the useful one at finite scales: it retains the crossover between the main and leakage lanes. The collapsed form is the version that can be inserted into an exponent ledger. Notice that the slower decay wins: if $\delta < \gamma$ and $\ell > 0$, the leakage term has exponent δ, not γ.

3 Margin and endpoint accounting

The signed four-packet interface carries an exact identity

$$m^2 = \frac{D}{G} m_D^2, \tag{6}$$

where m_D is the diagonal margin and m is the margin after reassembly. Equation (6) is an interface assumption for this compiler, not a new arithmetic estimate.

Corollary 2 (margin compiler)**.** *Under the hypotheses of Theorem 1 and* (6)*,*

$$m^2 \ge \frac{m_D^2}{BX^{-\gamma} + c_\ell X^{-\delta}}. \tag{7}$$

If, in addition, $m_D \ge cX^{-\eta_D - \varepsilon}$ with $c > 0$, then

$$m \ge cC^{-1/2} X^{-\eta_D + \kappa/2 - \varepsilon}. \tag{8}$$

In a ledger which records only nonnegative losses, one may use

$$\eta_{\text{eff}} = \max(0, \eta_D - \kappa/2).$$

Proof. Substitute the gain bound from Theorem 1 into (6). Taking square roots after using (3) gives (8). If the exponent of X is positive, replacing the corresponding negative loss by zero weakens the lower bound for $X \ge 1$. □

Suppose the remaining scalar lane has the inherited form $|S| \leq AX^{E_0-\sigma+\varepsilon}$. The endpoint ledger then receives the effective saving $\sigma - \eta_{\mathrm{eff}}$, so the strict target is

$$\sigma - \eta_{\mathrm{eff}} > \frac{1}{400}. \tag{9}$$

The strict inequality is essential: equality is a borderline unpaid case. The constant $C^{-1/2}$ affects constants, while κ controls the power lane.

4 Sharpness and the leakage obstruction

Proposition 3 (sharpness under the information model)**.** *For any admissible $X, d, B, \ell, a, \gamma, \delta$ with $C > 0$, the formal choices*

$$D = dX^a, qquadG = BX^{-\gamma}D + \ell X^{a-\delta}$$

satisfy the hypotheses and make (2) *and* (4) *equalities. If $\delta < \gamma$ and $\ell > 0$, the gain has asymptotic order X^δ up to constants, so a universal exponent γ conclusion is false under these inputs.*

Proof. Substitution gives $D = dX^a$ and $G/D = BX^{-\gamma} + c_\ell X^{-\delta}$ exactly. When $\delta < \gamma$, the ratio of the main term to the leakage term tends to zero, and the latter determines the order of the sum. $\square$

Thus the compiler identifies a precise failure mode for a tempting shortcut: one may not normalize an additive remainder by an upper or nominal source scale. Its actual payment is determined by the available lower bound on D. The equality family is an information-model adversary, not a claim that the literal TPC source realizes equality.

5 Exact fixtures and parent transfer

The release certificate uses rational arithmetic throughout. Table 1 shows representative normalized bounds; "two" is the exact sum in (2), while "collapsed" is the coarser right side of (3).

case	X	B	ℓ/d	γ	δ	two / collapsed
balanced	8	1/2	1/3	2	3	13/1536 / 5/384
slow leak	16	1	2	4	1	8193/65536 / 3/16
equal exponents	10	3/5	1/5	2	2	1/125 / 1/125
no leakage	9	7/4	0	3	5	7/2916 / 7/2916
leakage only	32	0	1/2	5	1	1/64 / 1/64
fractional constants	25	2/3	4/5	1	3	6262/234375 / 22/375

Table 1: Exact rational two-term and collapsed normalized bounds.

The first two rows make the bottleneck visible: in the slow-leakage row the nominal main exponent is 4, but the certified collapsed exponent is 1. The leakage-only row is an exact no-main-term adversary. Four margin fixtures replay (7); their two-term lower bounds are never smaller than the collapsed lower bounds. Four endpoint fixtures classify strict, leakage-limited, borderline, and unpaid cases, including the exact equality in (9).

As a separate provenance check, the twelve TPC-279 parent rows are copied only in their already-certified (q, Δ) coordinates. The transfer preserves eight positive-deficit and four negative-deficit rows. This finite census validates the coordinate interface; it does not assert that the literal source has an additive leakage decomposition or a growing deficit.

6 Route consequence and claim firewall

The new result is a conditional compiler, not an arithmetic payment. To use it on the twin-prime route one still needs a literal estimate of the form (1), including a source lower bound and a typed identification of the leakage term. In particular, the theorem does not convert a finite table of favorable gains into a power estimate.

claim	status
two-term normalization and gain	PROVED CONDITIONAL
dominant exponent $\min(\gamma, \delta)$	PROVED CONDITIONAL
equality-family sharpness	PROVED CONDITIONAL
finite rational fixtures and parent transfer	NUMERICALLY CERTIFIED
literal growing source decomposition	OPEN
arithmetic L^2 / full Gate B	OPEN
fixed-power credit / twin-prime result	0 / NONE

The Session-named Route A/Route B evaluator files are absent from the checkout; the project proof package, exact certificate, independent checker, stress audit, and fail-closed Bridge-B checker supply the scoped local evaluation.

References

[1] Liang Wang. A minimal coherence-to-gain criterion for four-packet reassembly, 2026. TPC-279 project release, 2026.